concept. The representation is merely a direction because $\alpha\bar{\ell}_W$ is also a linear representation for any $\alpha > 0$ (i.e., there is no notion of magnitude). In the case of concepts corresponding to counterfactual pairs of words, this direction can be shown to be proportional to the "linear probing" direction, and proportional to $g(Y(1)) - g(Y(0))$ for any counterfactual pair $Y(1), Y(0)$ that differ on W.

3 General Concepts and Hierarchical Structure

Our high-level strategy will be to build up from binary concepts to more complex structure. We begin by defining the basic building blocks.

Binary and Categorical Concepts The most general concept we address in this paper is a categorical concept, which refers to any concept corresponding to a categorical latent variable. This includes binary concepts as a special case. We consider two kinds of binary concept: *binary features* and *binary contrasts*. A binary feature $W \in_R \{\texttt{not_w}, \texttt{is_w}\}$ is an indicator of whether the output has the attribute w. For example, if the feature `is_animal` is true, then the output will be about an animal. A binary contrast $\texttt{a} \Rightarrow \texttt{b} \in_R \{a, b\}$ is a binary variable that contrasts two specific attribute values. For example, the variable $\texttt{mammal} \Rightarrow \texttt{bird}$ is a binary contrast. In the particular case where the binary contrast can correspond to counterfactual pairs of words (e.g., $\texttt{male} \Rightarrow \texttt{female}$), the concept matches the definition used in Park et al. (2024).

Hierarchical Structure The next step is to define what we mean by a hierarchical relation between concepts. To that end, to each attribute w, we associate a set of tokens $\mathcal{Y}(w)$ that have the attribute. For example, $\mathcal{Y}(\texttt{mammal}) = \{\text{“ dog”}, \text{“ cats”}, \text{“ Tiger”}, \dots\}$. Then,

Definition 2. A value z is *subordinate* to a value w (denoted by $z \prec w$) if $\mathcal{Y}(z) \subseteq \mathcal{Y}(w)$. We say a categorical concept $Z \in_R \{z_0, \dots, z_{n-1}\}$ is subordinate to a categorical concept $W \in_R \{w_0, \dots, w_{m-1}\}$ if there exists a value w_Z of W such that each value z_i of Z is subordinate to w_Z.

For example, the binary contrast $\texttt{dog} \Rightarrow \texttt{cat}$ is subordinate to the binary feature $\{\texttt{is_mammal}, \texttt{not_mammal}\}$, and the binary contrast $\texttt{parrot} \Rightarrow \texttt{eagle}$ is subordinate to the categorical concept $\{\texttt{mammal}, \texttt{bird}, \texttt{fish}\}$. On the other hand, $\texttt{dog} \Rightarrow \texttt{eagle}$ is not subordinate to $\texttt{bird} \Rightarrow \texttt{mammal}$, and $\texttt{bird} \Rightarrow \texttt{mammal}$ and $\texttt{live_in_house} \Rightarrow \texttt{live_in_water}$ are not subordinate to each other.

Linear Representations of Binary Concepts Now we return to the question of how binary concepts are represented. A key desideratum is that if $\bar{\ell}_W$ is a linear representation, then moving the context embedding in this direction should modify the probability of the target concept *in isolation*. If adding $\bar{\ell}_W$ also modified off-target concepts, it would not be natural to identify it with the target concept W. In Definition 1, this idea is formalized by the requirement that the probability of causally separable concepts is unchanged when the representation is added to the context.

We now observe that, when there is hierarchical structure, this requirement is not strong enough to capture 'off-target' behavior. For example, if $\bar{\ell}_{\texttt{animal}}$ captures the concept of animal vs not-animal, then moving in this direction should not affect the relative probability of the output being about a mammal versus a bird. If it did, then the representation would actually capture some amalgamation of the animal and mammal concepts. Accordingly, we must strengthen our definition:

Definition 3. A vector $\bar{\ell}_W$ is a linear representation of a binary concept W if

$$\mathbb{P}(W = 1 \mid \ell + \alpha\bar{\ell}_W) > \mathbb{P}(W = 1 \mid \ell), \text{ and} \tag{3.1}$$

$$\mathbb{P}(Z \mid \ell + \alpha\bar{\ell}_W) = \mathbb{P}(Z \mid \ell), \tag{3.2}$$

for all contexts ℓ, all $\alpha > 0$, and all concept variables Z that are either subordinate to or causally separable with W. Here, if W is a binary feature for an attribute w, then $W = 1$ denotes $W = \texttt{is_w}$.

Notice that, in the case of binary contrasts defined by counterfactual pairs, this definition is equivalent to Definition 1, because such variables have no subordinate concepts.

4 REPRESENTATIONS OF COMPLEX CONCEPTS

We now turn to how complex concepts are represented. The high-level strategy is to show how to represent binary features as vectors, show how geometry encodes semantic composition, and then use this to construct representations of complex concepts.

4.1 VECTOR REPRESENTATIONS OF BINARY AND CATEGORICAL CONCEPTS

To build up to complex concepts, we need to understand how to compose linear representations of binary features. At this stage, the representations are only *directions* in the representation space—they do not have a natural notion of magnitude. In particular, this means we cannot use vector operations (such as addition) to compose representations. To overcome this, we now show how to associate a magnitude to the linear representation of a binary feature. The key is the following result, which connects binary feature representations and word unembeddings:

Theorem 4 (Magnitudes of Linear Representations)**.** *Suppose there exists a linear representation (normalized direction) $\bar{\ell}_W$ of a binary feature W for an attribute w. Then, there is a constant $b_w > 0$ and a choice of unembedding space origin $\bar{\gamma}_0^w$ in eq.* (2.2) *such that*

$$\begin{cases} \overline{\ell_W^\top} g(y) = b_w & \text{if } y \in \mathcal{Y}(w) \\ \overline{\ell_W^\top} g(y) = 0 & \text{if } y \notin \mathcal{Y}(w). \end{cases} \tag{4.1}$$

Further, if there exist d attributes $\{w_0, \ldots, w_{d-1}\}$ such that the linear representations of the binary features for these attributes are linearly independent, we can choose a canonical origin $\bar{\gamma}_0$ in eq. (2.2)*.*

All proofs are given in Appendix B.

The theorem says that, if a (perfect) linear representation of the `animal` feature exists, then every token having the animal attribute has the *same* dot product with the representation vector; i.e., "cat" is exactly as much `animal` as "dog" is. If this weren't true, then increasing the probability that the output is about an animal would also increase the relative probability that the output is about a dog rather than a cat. In practice, such exact representations are unlikely to be found by gradient descent in LLM training. Rather, we expect $\overline{\ell_W^\top} g(y)$ to be isotropically distributed around b_w and 0, with variances that are small compared to b_w (so that animal and non-animal words are well-separated.)

With this result in hand, we can define a notion of vector representation for **binary features**:

Definition 5. We say that a binary feature W for an attribute w has a *vector representation* $\bar{\ell}_w \in \mathbb{R}^d$ if $\bar{\ell}_w$ satisfies Definition 3 and $\|\bar{\ell}_w\|_2 = b_w$ in Theorem 4. If the vector representation of a binary feature is not unique, we say $\bar{\ell}_w$ is the vector representation that maximizes b_w.

We have now moved from representations as directions to representations as vectors. Definition 5 and Theorem 4 give a simple way of composing binary features into **binary contrasts**:

Corollary 6 (Binary Contrasts Are Vector Differences of Binary Features)**.** *Let $w_0 \Rightarrow w_1$ be a binary contrast, and suppose there exist vector representations $\bar{\ell}_{w_0}$ and $\bar{\ell}_{w_1}$ for each attribute w_0 and w_1. Then, the difference $\bar{\ell}_{w_1} - \bar{\ell}_{w_0}$ is a linear representation $\bar{\ell}_{w_0 \Rightarrow w_1}$ in the sense of Definition 3.*

We can also apply ordinary vector space operations to construct representation of **categorical concepts**, e.g., $\{$`mammal`, `reptile`, `bird`, `fish`$\}$. There is now a straightforward way to define the representation of such concepts:

Definition 7. The *polytope representation* of a categorical concept $W = \{w_0, \ldots, w_{k-1}\}$ is the convex hull of the vector representations of the elements of the concept.[3]

In Appendix D, we state and prove a generalization of Corollary 6 to categorical concepts (which proves Corollary 6 itself).

[3]In Appendix C, we prove that the polytope representation for a "natural" categorical concept is a simplex.

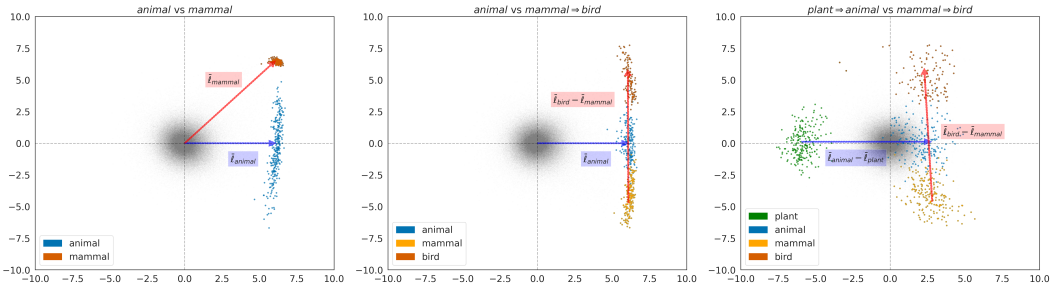

Figure 2: Hierarchical semantics are encoded as orthogonality in the representation space (Theorem 8). The plots show the projection of the unembedding vectors onto 2D subspaces: $\text{span}\{\bar{\ell}_{\texttt{animal}}, \bar{\ell}_{\texttt{mammal}}\}$ (left; (a)), $\text{span}\{\bar{\ell}_{\texttt{animal}}, \bar{\ell}_{\texttt{bird}} - \bar{\ell}_{\texttt{mammal}}\}$ (middle; (b)), and $\text{span}\{\bar{\ell}_{\texttt{animal}} - \bar{\ell}_{\texttt{plant}}, \bar{\ell}_{\texttt{bird}} - \bar{\ell}_{\texttt{mammal}}\}$ (right; (c)). Gray points indicate all 256K tokens in the vocabulary, and the colored points are the tokens in $\mathcal{Y}(w)$. The blue and red vectors are used to span the 2D subspaces.

4.2 Hierarchical Orthogonality

Now, we turn to the question of how hierarchical relationships between concepts are encoded in the representation space. The core intuition is that manipulating the "animal" concept should not affect relative probabilities of the "mammal" and "bird" concepts, so we might expect the representations of `animal` and `mammal` $\Rightarrow$ `bird` to be orthogonal. The following result formalizes this intuition by connecting the vector and semantic structures. The result is illustrated in Figure 2.

Theorem 8 (Hierarchical Orthogonality)**.** *Suppose there exist the vector representations for all the following binary features. Then, we have that*

(a) $\bar{\ell}_w \perp \bar{\ell}_z - \bar{\ell}_w$ *for* $z \prec w$*;*

(b) $\bar{\ell}_w \perp \bar{\ell}_{z_1} - \bar{\ell}_{z_0}$ *for* $Z \in_R \{z_0, z_1\}$ *subordinate to* $W \in_R \{\texttt{not_w}, \texttt{is_w}\}$*;*

(c) $\bar{\ell}_{w_1} - \bar{\ell}_{w_0} \perp \bar{\ell}_{z_1} - \bar{\ell}_{z_0}$ *for* $Z \in_R \{z_0, z_1\}$ *subordinate to* $W \in_R \{w_0, w_1\}$*; and*

(d) $\bar{\ell}_{w_1} - \bar{\ell}_{w_0} \perp \bar{\ell}_{w_2} - \bar{\ell}_{w_1}$ *for* $w_2 \prec w_1 \prec w_0$*.*

Together, Definition 7 and Theorem 8 give the simple structure illustrated in Figure 1: hierarchical concepts are represented as direct sums of polytopes. This direct sum structure is immediate from Theorem 8. We emphasize that all of the results—involving differences of representations—are only possible because we have *vector* representations (mere directions would not suffice).

5 Experiments

We now turn to empirically testing the theoretical results. Here, we present our empirical results on the Gemma-2B model (Mesnard et al., 2024).[4] In Appendix F, we further present empirical results on the LLaMA-3-8B model (Dubey et al., 2024), for which our findings are largely analogous.

5.1 Setup

Canonical Representation The results in this paper rely on transforming the representation spaces so that the Euclidean inner product is a causal inner product, aligning the embedding and unembedding representations. Following Park et al. (2024), we estimate the required transformation as:

$$g(y) = \text{Cov}(\gamma)^{-1/2}(\gamma(y) - \mathbb{E}[\gamma]) \tag{5.1}$$

where γ is the unembedding vector of a word sampled uniformly from the vocabulary. Centering by $\mathbb{E}[\gamma]$ is a reasonable approximation of centering by $\bar{\gamma}_0$ defined in Theorem 4 because this makes the projection of a random $g(y)$ on an arbitrary direction close to 0. This matches the requirement that the projection of a word onto a concept the word does not belong to should be close to 0.

[4]Code is available at github.com/KihoPark/LLM_Categorical_Hierarchical_Representations.